

Lab 2 instructions — Explaining a Model with SHAP (OPIM 5512)

Module 2 • keep this open during the lab. Every click you need tonight, in order. The model is already trained. Tonight is about **explaining** it — and doing it through the **GitHub workflow** you know: branch → commit → push → pull request → review → merge.

 Read [SHAP_LECTURE.md](#) **first** (or follow along live). Your only coding task is **one SHAP line**. Type it.

Where every file goes

Grey = the repo's structure & given data. Purple = the plots and words YOU add tonight.

```
your-repo/
|-- README.md           data dictionary + SHAP lecture (given)
|-- REPORT.md           you fill the arrow lines at the end
|-- data/energy_model_data.csv  (given)
|-- images/             <-- drag all four PNGs here
|   |-- importances_builtin.png  (A • given cell)
|   |-- shap_global.png         (A • you make)
|   |-- predicted_vs_actual.png  (B • given cell)
|   |-- shap_local.png          (B • you make)
`-- notebooks/           <-- Colab File > Save lands here
    |-- Lab2_A_Global_SHAP.ipynb  (A)
    |-- Lab2_B_Local_SHAP.ipynb   (B)
    |-- Lab2_Joint_Optional.ipynb
```

How things get there

Notebook > notebooks/

In Colab: File > Save (plain Save; not Ctrl+S).
Repo = yours | branch = dev-global / dev-local
Keep the same path every save.

PNGs > images/

Run all → the given plot saves itself; your SHAP line saves the other.
Run the download cell → both land in Downloads.
GitHub Desktop > Repository > Show in Explorer → drag into images/
A (1) in the name? Rename it back — the report links to exact names.

Then, in GitHub Desktop

Top bar must say your dev- branch (not main).
Commit with a real message > Push >
github.com: Compare & pull request > partner approves.

Before we start — check three things

1. **GitHub Desktop opens and you're signed in.**
2. **colab.research.google.com opens** and you're signed into a Google account.
3. You remember the Lab 1 loop. Same moves tonight; only the plots change.

Pick roles: Partner A owns **global** (what drives the model overall). Partner B owns **local** (why one specific prediction). You explain the *same* model two ways, then merge.

Part 1 — Set up the shared repo (Partner A drives, ~10 min)

1.1 Make the repo from the template

Open <https://github.com/drddave-teaching/opim5512-lab2-template> → green **Use this template** → **Create a new repository**.

Field	What to put
Owner	you (your personal account)
Repository name	opim5512-lab2-<netidA>-<netidB>
Visibility	Public (the notebooks read the data over a raw link; nothing secret)

Click **Create repository**. You now have a repo with the data, both notebooks, an empty `images/`, a README with the data dictionary, and a `REPORT.md` skeleton. **You built none of it.**

1.2 Add your partner

Repo **Settings** → **Collaborators** → **Add people** → Partner B's GitHub username → **Add**. **Partner B:** accept the invite (email, or the 🔔 on github.com). Until you accept, you can't push.

1.3 Protect `main`

Repo **Settings** → **Rules** → **Rulesets** → **New ruleset** → **New branch ruleset**: - Name: `protect main` · Enforcement: **Active** - Target branches → **Add target** → **Include default branch** - Rules: ☒ **Require a pull request before merging** → **Required approvals: 1** → **Create**

The two-person gate: nothing lands on `main` without your partner's approval, and **you can't approve your own PR**. (*Rehearsing solo with one account? Set Required approvals to 0.*)

1.4 Both partners: clone it — once

GitHub Desktop → **File** → **Clone repository** → **GitHub.com tab** → find the repo (🔄 refresh if needed) → note the **Local path** → **Clone**.

This is the *only* clone. From here, **Pull** brings work down, **Push** sends yours up. The **Changes** panel shows *what changed*; to see the files, use **Repository** → **Show in Explorer**.

1.5 Each partner: make your branch

Current branch → **New branch** → `dev-global` (A) / `dev-local` (B) → **Create branch** → **Publish branch**.

● Read the top bar before every commit. If it says `main`, stop and switch.

Part 2 — Explain the model (each partner, on your own branch, ~30 min)

2.1 Open *your* notebook in Colab, from *your* repo

colab.research.google.com → **File** → **Open notebook** → **GitHub tab** → paste your repo URL → open `notebooks/Lab2_A_Global_SHAP.ipynb` (A) or `notebooks/Lab2_B_Local_SHAP.ipynb` (B). Open in its **own tab**.

2.2 Run the setup (given)

Runtime → **Run all**. The first cell runs `!pip install shap`, loads the data, and fits the model (prints R^2 — that's how good it is). The SHAP setup cell then builds `shap_values` — one push in MW per feature, per row. **You didn't**

write any of that; you're about to use it.

You also get one plot for free: - **Partner A:** the model's **built-in importances** bar chart → saves `importances_builtin.png`. - **Partner B:** the **predicted-vs-actual** scatter → saves `predicted_vs_actual.png`.

2.3 Write your one SHAP line (the thing you code)

In the **TODO** cell, the exact shape is printed right above it.

Partner A — the beeswarm (global):

```
shap.plots.beeswarm(shap_values, show=False)
plt.gcf().savefig("shap_global.png", dpi=150, bbox_inches="tight"); plt.close()
```

Then **look at it:** which feature is #1? Do high (red) values push demand up or down? Did SHAP reorder the built-in ranking?

Partner B — the waterfall (local):

```
i = int(np.argmax(model.predict(X))) # the hour the model predicts highest
print("explaining:", df.loc[i, "hour"], "| actual:", round(df.loc[i, "load_mw"]), "MW")
shap.plots.waterfall(shap_values[i], show=False)
plt.gcf().savefig("shap_local.png", dpi=150, bbox_inches="tight"); plt.close()
```

Then **look at it:** which features pushed this hour **up** (red) and which pulled it **down** (blue)? Hour of day, weather, or both? That sentence is your finding.

Keep the filenames **exactly** as given — `REPORT.md` links to them.

2.4 Download your two PNGs

Run the **download** cell. Two files land in **Downloads**. If one "isn't found yet," run the cell that makes it, then re-run.

2.5 Save the notebook back to GitHub (File → Save)

Colab **File** → **Save**. (No "Save a copy in GitHub" item — plain **Save** commits because you opened it *from* GitHub. **Ctrl+S alone only autosaves to Drive**.) In the dialog: 1. **Repository** — scroll to **your** repo (long list, not searchable). 2. **Branch** — your `dev-` branch. 3. **File path** — leave it as `notebooks/Lab2_A_Global_SHAP.ipynb` (or `B_Local...`). **Same path every time**. 4. **Commit message** — a real one: `add global SHAP beeswarm / add local SHAP waterfall`.

💡 Each save is a snapshot. Edit again → save again.

2.6 Drag the PNGs into the repo

GitHub Desktop → **Repository** → **Show in Explorer** → drag both PNGs from **Downloads** into `images/`.

⚠️ A (1) in a filename means you downloaded twice — **rename it back** to the exact name first.

2.7 Commit and push

Back in **GitHub Desktop**: confirm the top bar says **your dev- branch**. **Summary** = `add global SHAP plots` (or `add local SHAP plots`) → **Commit** → **Push origin**. If it says **Pull origin** first (your Colab save coming down), click that, then Push.

Part 3 — Review & merge (both, ~20 min)

3.1 Open your pull request

github.com shows a yellow bar: **Compare & pull request** → check it's `dev-global` → `main` (or `dev-local`) → title it → **Create pull request** → **Reviewers** → your partner.

3.2 Review your partner's

Open your **partner's** PR → **Files changed**. Read their SHAP cell and look at the PNG. Does the plot answer their question — global "what matters," local "why this hour"? **Review changes** → **Approve** → **Submit review**. Leave one real comment.

3.3 Merge, delete, pull

- **Merge pull request** → **Confirm** → **Delete branch**. Both PRs.
- **Both**: GitHub Desktop → **Current branch** → `main` → **Fetch origin** → **Pull origin**. Open `images/` : **four PNGs** — the global view and the local view of the same model, side by side.

Part 4 — The report (one screen, two people, ~20 min)

4.1 Fill in `REPORT.md`

github.com → `REPORT.md` →  **Edit**. The four plots already render. Replace each → line with **one sentence** — every number gets a unit (MW, °F, hour). The key sentence to nail: **does global SHAP and the local hour tell the same story about what drives demand?** Add one honest line under *What SHAP can't tell us* (hint: SHAP explains the model, not physical cause).

4.2 It goes through a pull request too

Commit changes... → **Create a new branch** `report` → **Propose changes** → **Create pull request** → the *other* partner **approves** → **Merge** → delete branch. `main` is protected — that's the rule working.

4.3 If you're ahead: the dependence plot

Open `notebooks/Lab2_Joint_Optional.ipynb` → run → download `shap_dependence.png` → drag into `images/` → commit on a branch → PR → merge. It drops into section 5. Chase the question at the bottom of that notebook.

4.4 Read-out

Two plots on the screen — **beeswarm** (what the model leans on) and **waterfall** (why one hour) — one sentence each. Then **Insights** → **Network** shows the loop going both ways.

When it breaks

Symptom	Fix
No module named 'shap'	The setup cell installs it. Runtime → Run all from the top; don't skip cell 1.
Beeswarm/waterfall cell errors on shap_values	You skipped the SHAP setup cell. Run all from the top.
Push rejected on main	You're on <code>main</code> . Switch to your <code>dev-</code> branch and commit there — the rejection is the protection working.
"Where's Save a copy in GitHub?"	It doesn't exist. Plain File → Save .
Saved but GitHub didn't change	You hit Ctrl+S (Drive autosave). File → Save , and re-save after each edit.
Report shows a broken image	Filename mismatch — usually a (1) in the PNG name, or it's in the wrong folder. Exact name, in <code>images/</code> .
Can't pick my branch in the Colab save dialog	It lists only <i>existing</i> branches. Make it in GitHub Desktop first (1.5), then save.

Never used this workflow? The Lab 1 kit walks every GitHub move slowly: [Lab1_FirstCommit](#).